

Chong-Kai Li

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EDUCATION

National Yang Ming Chiao Tung University

Sep 2024 – Aug 2026

Master's Program, Institute of Artificial Intelligence

National University of Tainan

Sep 2019 – Jul 2024

Bachelor's Program, Department of Computer Science and Information Engineering

SKILLS

Programming Languages: Python, C/C++, Java, SQL, HTML/CSS, MATLAB

Tools & Framework: PyTorch, Transformers, LangChain, scikit-learn, OpenCV, pandas, numpy, docker, Linux, Shell

AWARDS

Jun 2025 Outstanding Student Award

Dec 2024 Outstanding Student Award

Dec 2024 Mobile Communications Practice Competition - Honorable Mention

Dec 2023 Capstone Exhibition - 1st Place & Best Popularity Award

PROJECTS

Multi-Source Attack Path Reconstruction with Weighted Attack Graphs

Sep 2025 – Aug 2026

Developed a NICS-sponsored attack path reconstruction system integrating Zeek/NetFlow, Suricata NIDS, Wazuh HIDS, and VirusTotal CTI. Built a dual-stream MaliciousFlowTransformer + SecureBERT 2.0 model for event-level threat scoring and a multi-source weighted attack graph with reverse beam search to reconstruct time-ordered attack paths from known victim hosts. Deployed on TWCC with Docker Compose and achieved F1 scores of **1.0000** on DARPA 2000 and the TWCC custom testbed and **0.9730** on the full six-day ISCXIDS2012 dataset, outperforming recent state-of-the-art baselines on evaluated benchmark scenarios.

NYCU-Bot - Multi-Agent Award Announcement & Social Media Automation System

Sep 2025 – Dec 2025

Developed a multi-agent system that monitors NYCU's AI institute website for new award announcements and automatically publishes bilingual congratulatory posts across Twitter/X, Facebook, Instagram, LinkedIn, and Reddit. Designed a hierarchical agent architecture (an InformationAgent, Father/Mother coordinator agents, and platform-specific publishing agents) coordinated through an internal message bus, with a LangChain-based ContentAgent using a locally hosted Ollama DeepSeek-R1 7B model to generate localized captions and hashtags. Supported both one-shot execution and continuous polling with configurable check intervals.

RNN & Transformer Course - Intelligent Research Companion

Jun 2025

Developed a multi-agent academic paper analysis system using Mistral OCR, citation extraction, and automated reference downloads. Implemented hierarchical agents (orchestrator, coordinator, and task-specific agents) powered by LangChain, RAG, and local LLMs via Ollama. The system enables automated literature reviews and structured knowledge extraction, improving efficiency for academic research.

AI Module Course - Agrivoltaics Smart Micro-grid for Leisure Farms

Dec 2024

Designed an agrivoltaics system integrating solar power with IoT-enabled smart micro-grid. Features included real-time solar monitoring (MQTT), AI-based anomaly detection, automated LED lighting control, and an ML model predicting yields from environmental data and Kaggle datasets. Deployed the system into a web platform with LDR-based real-time sensing, achieving **Honorable Mention** in competition.

Senior Project - AI-based Video Analysis for Badminton Strategies

Dec 2023

Built a system to analyze badminton players' posture matches using YOLOv7 for shot detection and OpenPose for joint extraction. Trained models on multiple processed datasets to identify optimal prediction strategies. The project demonstrated multi-modal data fusion and won **1st place** and the **Popularity Award** in the Multimedia group at the university graduated exhibition.

INTERNSHIP EXPERIENCE

AI Engineer Intern

Jun 2025 – Aug 2025

- Developed an intelligent Q&A system for manufacturing operations, providing real-time solutions and contingency measures for factory floor inquiries.
- Implemented advanced fine-tuning techniques including RAG, SFT, and LoRA to optimize LLM performance for domain-specific manufacturing scenarios.
- Gained expertise in state-of-the-art fine-tuning methodologies including PPO, GRPO, and other contemporary RL approaches for modern LLM training.